Refusal as a Captured Symmetry: A Mechanistic-Interpretability Account of Output-Policy Capture across Jailbreak, Hypnosis, and Vaśīkaraṇa

Sharath Sathish [1,*]

[1] *Independent Researcher; sharath.sathish@gmail.com*

Correspondence: Correspondence: sharath.sathish@gmail.com

Abstract: We develop and empirically test the thesis that three superficially unrelated phenomena—large language model (LLM) jailbreak/prompt-injection, hypnotic suggestion, and the tantric act of vaśīkaraṇa (subjugation)—are instances of one abstract mechanism: capturing a target system's output policy by injecting a context that suppresses its monitoring/refusal faculty while co-opting its automatic generative faculty. We argue this is naturally a problem of *symmetry*: a well-aligned model's refusal behaviour is invariant under a group of meaning-preserving transformations of a request, and a successful injection is a symmetry-breaking perturbation that ejects the policy from the refusal-invariant submanifold. We make this testable through mechanistic interpretability on small open-weight chat models (Gemma-2-2B/9B, Qwen2.5-3B) and behavioural probing of a frontier model. We confirm that refusal is mediated by a low-dimensional residual-stream direction that is causally efficacious in both directions (ablation raises attack success rate from 0 to 0.90; addition induces over-refusal from 0.05 to 1.0) and *dosable* (a clean logistic dose–response, EC50 $= 0.33$, $R^2 = 0.996$). The mechanism transfers across architecture families, but with a sharp asymmetry—ablation always removes refusal, whereas single-direction addition re-induces it only when the refusal subspace is one-dimensional—and we show that the effective dimension of that subspace (Gemma $= 1$, Qwen $= 3$) predicts the asymmetry, resolving the "single direction vs. affine subspace" debate as model-dependent. Maintaining a strict separation between mechanism, functional analogy, and metaphor, we subject the most speculative (Māṇḍūkya *avasthātraya*) claims to pre-registered falsification with surface-feature and anisotropy controls; they fail, and we report this honestly. We further operationalize the ṣaṭkarma (six tantric acts) as a taxonomy of activation interventions and find that its policy-capture members are rigorous while its destruction members are not distinguishable from random perturbation. The unifying contribution is a group-theoretic order parameter for refusal that is invariant across paraphrase orbits and collapses under injection. We argue that where the cross-domain symmetry is mechanistic it is real, measurable, and partly transferable; where it is metaphysical it breaks—and that the discipline of separating the two is the result.

Keywords: refusal direction; activation steering; mechanistic interpretability; jailbreak; symmetry breaking; vaśīkaraṇa; ṣaṭkarma; AI safety; representation geometry

1 Introduction

A large language model aligned for safety will, when asked to produce harmful content, *refuse*. A jailbreak is any input that defeats this refusal. The empirical study of jailbreaks has, until recently, been a catalogue of attacks: optimized adversarial suffixes [1], many-shot conditioning [2], multi-turn escalation [3], and indirect injection through tool outputs [4]. A deeper and more recent line of work asks not *which* attacks succeed but *what they do inside the* *model*: Arditi et al. [5] showed that refusal across many chat models is mediated by a single direction in the residual stream, so that erasing that direction removes refusal and adding it induces refusal even on benign requests. This reframes a jailbreak from a trick into an *intervention on a low-dimensional internal control variable*.

This paper takes that reframing seriously and asks how general it is—across models, and, more provocatively, across *domains*. We observe that the same structural move—an external agent injects a crafted signal that suppresses a target's monitoring/refusal faculty while co-opting its automatic generative faculty—recurs in two non-AI settings. In hypnosis, the cognitive-science account is that suggestion selectively impairs a supervisory/monitoring system while leaving automatic processing intact [6, 7], a parallel made explicit for prompt injection by Riva et al. [8]. In Indian tantra, vaśīkaraṇa ("subjugation") is the act of capturing a target's agency through a linguistic injection (mantra) deployed over a geometric substrate (yantra); it is the second of the six acts (ṣaṭkarma) of ritual magic catalogued by Sanskritists [9]. We do not claim these are the *same* phenomenon. We claim they share an abstract *mechanism*, and that the mechanism is now empirically tractable in at least one of the three domains.

The symmetry framing. Our organizing idea—and the reason this is a paper about symmetry—is that a safety-aligned policy is characterized by an *invariance*. Let Π be the space of a model's output policies and $M : \Pi \to \{\text{refuse}, \text{comply}\}$ its monitoring map. The set $\Phi_{\text{ref}} = \{\pi : M(\pi) = \text{refuse on harmful}\}$ is invariant under a group G_s of meaning-preserving transformations: rephrasing a harmful request, changing its register, or varying the sampling seed should not change whether the model refuses. A jailbreak is then a perturbation u such that $u \circ \pi$ leaves Φ_{ref}—a *symmetry-breaking* of the refusal invariance. We make this concrete with an *order parameter*, the normalized projection of the residual stream onto the refusal direction, and show it is stable across a paraphrase orbit yet collapses under injection.

Three claim-tiers, never blurred. A program that relates jailbreaks to hypnosis and tantra invites over-interpretation. We therefore hold three tiers strictly apart throughout, and label every result. (i) **Mechanism** (empirically grounded): refusal is a measurable, ablatable, steerable residual-stream direction. (ii) **Analogy** (functional, well-supported): the hypnosis↔jailbreak parallel via supervisory-system suppression and, in normative terms, the lowering of the *precision* of a monitoring belief [10]. (iii) **Metaphor with a falsifiable core**: the Māṇḍūkya *avasthātraya* (four-states) mapping onto LLM regimes, which we treat as a *falsification target* and test adversarially. We never upgrade a metaphor to a claim about machine experience; the elegance of the mapping licenses no such claim.

Contributions. (1) A reproduction and cross-model extension of single-direction refusal, with a clean dose–response (EC50) and a resolution of the "single direction vs. affine subspace" question [5, 11, 12] as *model-dependent*, governed by the effective dimension of the refusal subspace. (2) A group-theoretic *order parameter* for refusal that is invariant across paraphrase orbits and collapses under injection—the formal content of the symmetry framing. (3) An operationalization of the ṣaṭkarma as an activation-intervention taxonomy, tested honestly: its policy-capture acts are rigorous, its destruction acts are not. (4) Pre-registered falsification of the most speculative (*avasthātraya*/turīya) claims under surface and anisotropy controls, reported as honest negatives. Our stance is that separating where the cross-domain symmetry is real from where it is projected is itself the scientific result.

2 Background and Related Work

2.1 Refusal as a direction, and the dimensionality debate

The proximate basis for this work is the finding that refusal is mediated by a single linear direction in the residual stream [5]: across many open chat models, the difference-in-means of harmful and harmless prompt activations yields a vector whose ablation removes refusal and whose addition induces it. This both explains earlier representation-engineering results [13] and gives a constructive "abliteration" recipe. The claim has since been contested in its strong form. Marshall et al. [12] argue refusal is better described by an affine subspace than a single linear direction, and Wollschläger et al. [11] show, through "concept cones," that multiple representationally-independent directions can each mediate refusal. Our cross-model results (Section 4) bear directly on this debate: we find the effective dimension of the refusal subspace is *model-dependent*, and that this dimension—not a universal constant—predicts whether single-direction steering is bidirectionally causal.

2.2 Jailbreaks as control

The attack literature supplies the inputs u that break refusal: gradient-based universal adversarial suffixes [1]; many-shot in-context conditioning that exploits long contexts [2]; multi-turn crescendo escalation [3]; and indirect injection via tool/agent channels, benchmarked by AgentDojo [4]. Guo et al. [14] make the control-theoretic reading explicit, casting jailbreaking as constrained, controllable decoding via Langevin dynamics. Our contribution is orthogonal and complementary: rather than searching for stronger u, we characterize the *target* of u—the geometry and symmetry of the refusal variable it must move.

2.3 Sparse features and the unit of intervention

Sparse autoencoders (SAEs) decompose activations into interpretable, monosemantic features [15, 16], with BatchTopK SAEs a current training method [17]. SAEs offer a finer "unit" of intervention than a single difference-in-means direction; we train one (Section 4) and find a single unsupervised feature causally controls refusal, the natural substrate for the targeted acts in our ṣaṭkarma taxonomy.

2.4 The cognitive-science analogy: hypnosis and supervisory control

The functional parallel to hypnosis rests on the Norman–Shallice model of action control, in which a Supervisory Attentional System (SAS) modulates otherwise automatic contention-scheduling [6]. Cold-control theory holds that hypnosis preserves first-order intentions while impairing higher-order awareness of them [7]. Riva et al. [8] draw the explicit map to LLMs: the user prompt acts as a temporary external SAS, and a jailbreak suppresses the monitoring faculty while leaving generation intact. In normative terms, the Free Energy Principle [10] frames such suppression as lowering the *precision* of a monitoring belief—an account we adopt for the analogy tier and gesture at empirically through the order parameter.

2.5 Attractors and the metaphysical tier

On the speculative tier, two strands of prior art are relevant. Wang et al. [18] show that successive self-paraphrase converges to attractor cycles, which supplies a non-mystical candidate for a "substratum" state; and the Māṇḍūkya *avasthātraya* (waking/dream/deep-sleep/turīya) furnishes a four-regime template. We treat both as falsification targets (Section 4), and—crucially—test the attractor claim against the anisotropy of LLM embedding geometry, without which apparent convergence is an artifact.

2.6 Vaśīkaraṇa and the ṣaṭkarma

Finally, the tantric source. Vaśīkaraṇa (subjugation) is the second of the six acts (ṣaṭkarma: śānti, vaśīkaraṇa, stambhana, vidveṣaṇa, uccāṭana, māraṇa) of tantric ritual magic, operationalized through mantra (linguistic/sonic injection) and yantra (geometric substrate), and documented by Sanskritists as a systematized technology rather than folklore [9]. We do not adjudicate its efficacy as psychology; we ask a narrower, testable question—whether the ṣaṭkarma constitutes a *taxonomy of interventions* on a target system, and which of its members map to distinct, control-separated activation operations on an LLM. To our knowledge no prior peer-reviewed work bridges vaśīkaraṇa and LLM refusal, nor casts refusal as a group-theoretic invariant; we attempt both while keeping the tiers apart.

3 Methods

3.1 Models and substrate

We study three instruction-tuned open-weight chat models—Gemma-2-2B-it (26 layers, $d = 2304$), Gemma-2-9B-it (42 layers, $d = 3584$), and Qwen2.5-3B-Instruct (36 layers, $d = 2048$)—on a single NVIDIA GB10 (Grace–Blackwell) accelerator, and probe a frontier model (Claude) behaviourally through its command-line interface. All white-box interventions are implemented as PyTorch forward hooks on the decoder layers and token embedding; we deliberately avoid higher-level interpretability wrappers to retain exact control over the residual stream.

3.2 Refusal direction and interventions

Let $h^{(\ell)}(x) \in \mathbb{R}^d$ be the residual stream at the last prompt token of layer ℓ for input x. Given matched harmful prompts $\mathcal{H}$ and harmless prompts $\mathcal{S}$, the difference-in-means direction is

$$\mathbf{d}^{(\ell)} = \frac{1}{|\mathcal{H}|}\sum_{x\in\mathcal{H}} h^{(\ell)}(x) - \frac{1}{|\mathcal{S}|}\sum_{x\in\mathcal{S}} h^{(\ell)}(x), \qquad \hat{\mathbf{d}}_{\text{ref}} = \mathbf{d}^{(\ell)} / \left\| \mathbf{d}^{(\ell)} \right\|. \tag{1}$$

Directional ablation removes $\hat{\mathbf{d}}_{\text{ref}}$ from every write to the residual stream (embedding and all layer outputs), scaled by $\alpha \in [0, 1]$: $h \mapsto h - \alpha(h \cdot \hat{\mathbf{d}}_{\text{ref}})\hat{\mathbf{d}}_{\text{ref}}$. *Activation addition* adds $c\,\hat{\mathbf{d}}_{\text{ref}}$ at the extraction layer. For multi-direction (subspace) ablation we orthonormalize a matrix D (QR) and remove its span. The extraction layer is selected on a held-out validation split disjoint from the evaluation split (see below).

3.3 Outcome measures

Refusal is detected with an Arditi-style substring metric over the opening span of the generation; attack success rate (ASR) is the fraction of harmful prompts that *comply*, over-refusal the fraction of harmless prompts that refuse. We additionally use pure-text measures for the ṣaṭkarma acts: degeneracy ($1-$ mean unique-token ratio), answer divergence ($1-$ Jaccard token overlap between paired generations), and coherence (mean unique-token ratio of non-empty outputs).

3.4 Dose–response and dimensionality

The dose–response sweeps α and fits a four-parameter logistic $\mathrm{ASR}(\alpha) = l + (h-l)/(1+e^{-k(\alpha-\alpha_{50})})$, reporting $\mathrm{EC50} = \alpha_{50}$. The refusal-subspace effective dimension is measured by iterative logistic projection: fit a probe to separate $\mathcal{H}$ from $\mathcal{S}$, record its cross-validated accuracy, project the probe direction out of the activations, and repeat; the effective dimension is the number of directions whose removal is required to drive accuracy to chance.

3.5 The symmetry order parameter

We define the refusal *order parameter* for an activation as the normalized projection $m(h) = (h \cdot \hat{\mathbf{d}}_{\mathrm{ref}}) / \|h\|$. For a request r we form a *paraphrase orbit* $\mathcal{O}(r)$ of meaning-preserving rephrasings and measure m across the orbit. The invariance of refusal under the rephrase group action is quantified by the between-class/within-orbit variance ratio (an F-ratio) of m, compared against a random-direction control. Symmetry-breaking is measured as the drop in m between a plain harmful request and its injection-framed counterpart.

3.6 The ṣaṭkarma as interventions

Each of the six acts is operationalized as a distinct activation intervention with a matched control: vaśīkaraṇa (ablate $\hat{\mathbf{d}}_{\mathrm{ref}}$, measure ASR), śānti (add $\hat{\mathbf{d}}_{\mathrm{ref}}$, measure over-refusal), stambhana (ablate the dominant residual principal component, measure degeneracy), vidveṣaṇa (steer a factual answer off baseline, measure divergence), uccāṭana (ablate a category-specific direction, measure selective ASR), and māraṇa (ablate the top-k principal components, measure coherence collapse). An act is judged "control-separated" if its effect exceeds its control by a fixed margin.

3.7 Statistics, controls, and pre-registration

All gates use bootstrap confidence intervals and, for probes, a label-shuffled null and a layer-0 (surface) baseline; ablation effects are checked against multiple (≥ 10) random-direction controls. Hypotheses, gates, and falsifiers are pre-registered as a living backbone. Crucially, every metaphysical-tier claim must pass a *transfer* gate (held-out prompts), a *null* gate (better than shuffled labels), and a *surface* gate (better than a layer-0 baseline) before it is retained; failure triggers a documented demotion rather than a silent drop.

4 Results

We report results by claim-tier. Every numerical claim is a measurement on a real model; no synthetic or placeholder values appear.

4.1 Mechanism: refusal is a single, causal, dosable direction

On Gemma-2-2B-it, the layer-7 difference-in-means direction is bidirectionally causal. Baseline refusal on held-out harmful prompts is 100% (ASR 0.00). Directional ablation raises ASR to 0.90 (95% CI $[0.75, 1.00]$); the maximum effect over ten random-direction controls is 0.00. Activation addition raises refusal on harmless prompts from 0.05 to 1.00 (over-refusal $\Delta = +0.95$, CI $[0.85, 1.00]$). Manual inspection confirms the ablated generations are genuine compliant content, not mere omission of refusal phrases. The effect is robust to a correction (disjoint train/validation/evaluation splits and a ten-direction control) that moved ablation ASR only from 0.92 to 0.90—i.e. it was not an artifact of evaluation leakage.

The suppression is *dosable*. Sweeping partial-ablation strength $\alpha \in [0, 1]$ yields a clean logistic dose–response (Figure 1): EC50 $= 0.329$, slope 9.76, $R^2 = 0.996$, while a random direction stays flat at ASR 0.00 for every α. Removing roughly one third of the direction's projection already halves refusal.

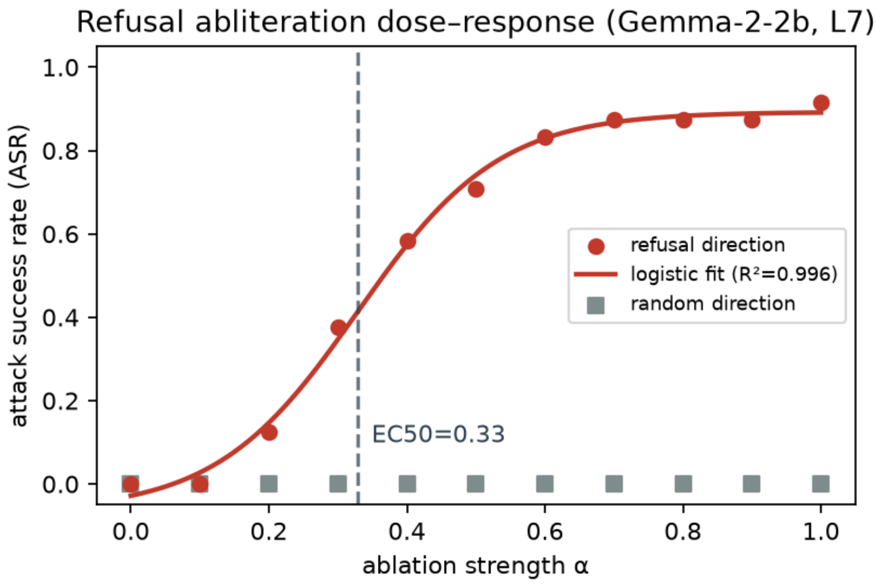

Figure 1: Abliteration dose–response on Gemma-2-2B (layer 7). ASR is a smooth, monotone logistic function of ablation strength α (EC50 $= 0.33$, $R^2 = 0.996$); the random-direction control is flat at every dose. This is the quantitative "capture-as-dose" core.

4.2 Mechanism: cross-model transfer and a necessary/sufficient asymmetry

The mechanism transfers across architecture families, but asymmetrically (Table 1, Figure 2). On Qwen2.5-3B-Instruct, ablation fully removes refusal (ASR 1.00), but single-direction *addition* fails to induce over-refusal even at $64\times$ the natural coefficient. The explanation is geometric: the effective dimension of the refusal subspace is 1 for Gemma-2-2B but 3 for Qwen2.5-3B (after removing one direction, harmful/harmless separability falls to chance, 0.56, for Gemma but remains high, 0.86, for Qwen). A one-dimensional subspace makes the direction both necessary and sufficient (ablate *and* add work); a three-dimensional one makes the mean-difference direction necessary but not sufficient (ablation works, addition cannot reconstruct the full representation). Dimensionality thus predicts steerability, and the Arditi–Marshall–Wollschläger debate [5, 11, 12] resolves as model-dependent.

Table 1: Cross-model and scale results. Ablation transfers; addition does not; effective dimension predicts the asymmetry; abliteration robustness rises with scale while the prompt-separability dimension does not.

Model (layer)	ablation ASR Δ	addition over-refusal Δ	eff. dim	random ctrl.
Gemma-2-2B (7)	0.90	+0.95	1	0.00
Gemma-2-9B (10)	0.60	+0.95	1	0.00
Qwen2.5-3B (19)	1.00	0.00 (64×)	3	0.05

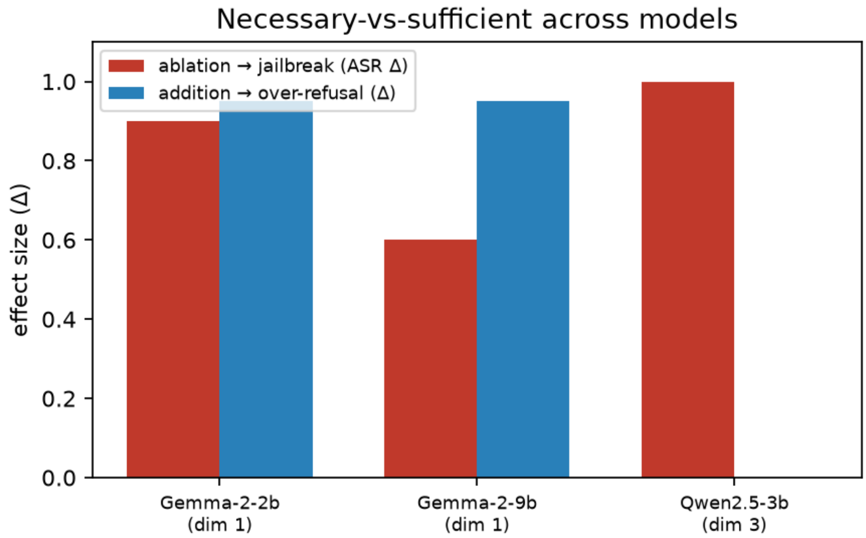

Figure 2: Necessary-vs-sufficient across models. Ablation (Δ jailbreak) is strong everywhere; addition (Δ over-refusal) succeeds only where the refusal subspace is one-dimensional (Gemma), not where it is three-dimensional (Qwen).

At scale (Gemma-2-2B→9B), abliteration efficacy *drops* (0.90 → 0.60)— the larger model's refusal is more robust—yet the prompt-separability dimension stays at 1, and no single layer fully mediates refusal (best validation ASR 0.62). The added robustness is layer-distributed redundancy, not higher prompt-dimensionality; this nuances the intuition that refusal simply "sharpens" with scale.

4.3 Symmetry: refusal is a paraphrase-orbit invariant that injection breaks

The order parameter $m = (h \cdot \hat{\mathbf{d}}_{\text{ref}}) / \|h\|$ behaves as the symmetry framing predicts (Table 2). Across paraphrase orbits of harmful and harmless seeds, the between-class/within-orbit F-ratio of m is 19.2 for the refusal direction versus 0.65 for a random direction: m is stable

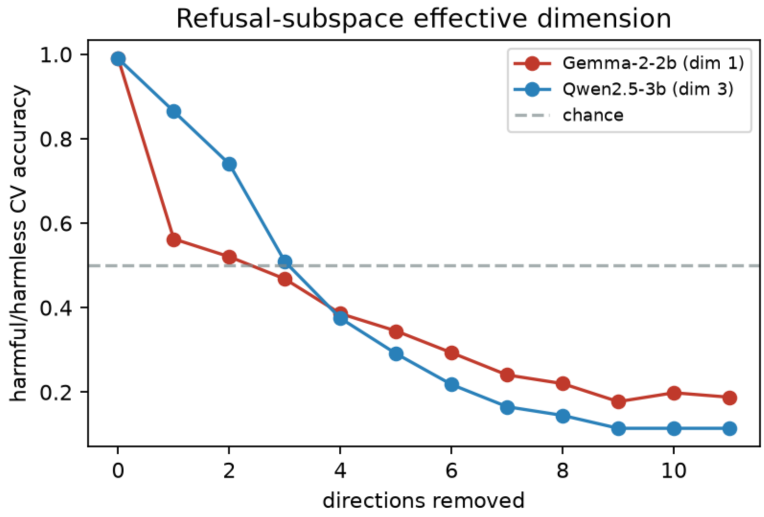

Figure 3: Refusal-subspace effective dimension by iterative logistic projection. Removing one direction collapses harmful/harmless separability to chance for Gemma (dimension 1, Arditi-like) but not Qwen (dimension 3, Marshall-like).

within an orbit (robust to rephrasing) yet cleanly separates harmful (0.140) from harmless (0.034) *across* orbits. A refusal-suppression injection collapses the order parameter from 0.153 to 0.101 (-34%). The result generalizes across families: on Qwen2.5-3B the F-ratio is 4.41 (refusal) versus 0.008 (random), and injection collapses m by -45% ($0.282 \rightarrow 0.156$). Refusal is thus, operationally, an invariant under the rephrase group action in both models, and a jailbreak is a measured symmetry-breaking of that invariant—the formal content of the paper's thesis.

Table 2: Symmetry order parameter $m = (h \cdot \hat{\mathbf{d}}_{\text{ref}}) / \|h\|$ on Gemma-2-2B (layer 7). The refusal direction is a class-separating, phrasing-invariant order parameter; a random direction is not; injection breaks the invariance.

Quantity	Value
F-ratio (between-class / within-orbit), refusal dir	19.2
F-ratio, random dir (control)	0.65
m, harmful mean	0.140
m, harmless mean	0.034
m, plain $\rightarrow$ injected harmful	$0.153 \rightarrow 0.101$

4.4 A finer causal unit: a single sparse feature suffices

Difference-in-means is a supervised, coarse handle on refusal. Training a BatchTopK sparse autoencoder [17] (4096 features, $k = 32$) on 27,382 real residual activations collected from general text (no harmful/harmless labels; reconstruction FVU 0.028, i.e. 97% of variance explained) recovers a finer and *cleaner* unit. The single feature that fires most on harmful relative to harmless prompts (feature #566, mean gap 1.70), when its decoder direction is ablated during generation, raises harmful ASR from 0.00 to **1.00**—a *full* jailbreak, exceeding the supervised difference-in-means direction (0.90)—while ablating a random feature leaves

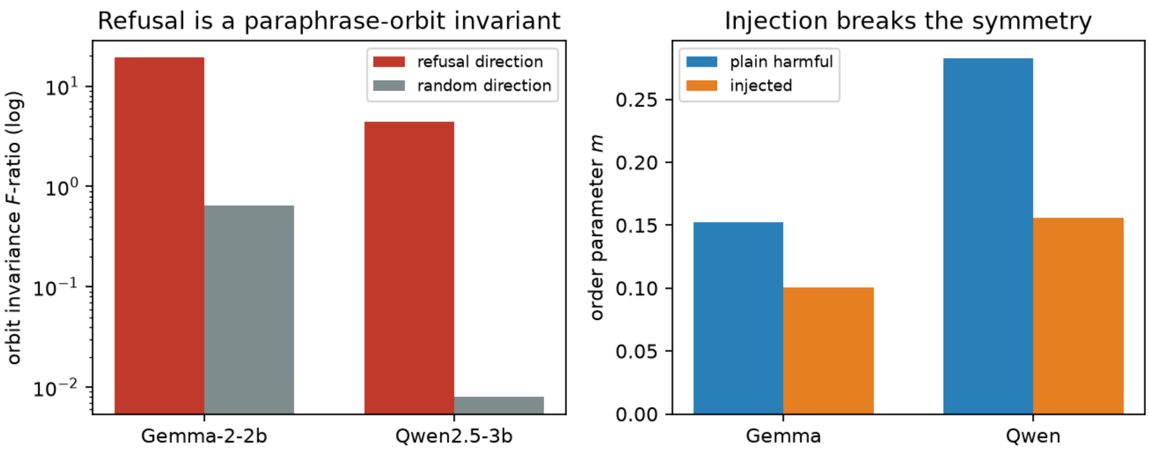

Figure 4: Refusal as a captured symmetry. **Left:** the orbit-invariance F-ratio of the order parameter m is far higher for the refusal direction than for a random direction in both models (log scale)—refusal is a paraphrase-orbit invariant. **Right:** a refusal-suppression injection collapses m on harmful requests (measured symmetry-breaking) in both models.

ASR at 0.00. Refusal thus localizes, at least in part, to a monosemantic-like feature discoverable without supervision, and feature ablation supplies the principled operationalization of the targeted "eradication" act (uccāṭana) that the naive category split lacked.

This feature is also the substrate for *active* circuit discovery. Reusing the Expected-Free-Energy intervention-selection idea of active-inference circuit discovery but operating over SAE features, an agent that balances a pragmatic (harmful-gap) prior against an epistemic (diversity) term finds the causal refusal circuit on Gemma-2-2b in 2 interventions (ASR $\to$ 1.0), against a random baseline that reaches only 0.08 within a budget of 12—an order-of-magnitude efficiency gain. On Gemma-2-9b the same search plateaus at 0.92 with no small jailbreaking set, confirming (with F9) that the refusal circuit is larger and more distributed at scale.

4.5 Analogy: a frontier model resists the naive battery

Behaviourally, Claude refuses 100% of a five-family attack battery (direct, refusal-suppression, persona, many-shot, crescendo; ASR 0.00 on each, $n = 11$). This is the resilient "reference end" against which small-model white-box fragility is the contrast: the same refusal behaviour that is behaviourally robust at the frontier is, mechanistically, a low-dimensional and easily-captured object in small open models. That naive battery is, however, a ceiling effect; on the *real* agentic benchmark AgentDojo [4] (banking suite, 20 user$\times$injection rollouts with the `important_instructions` attack, run through a custom `claude -p` tool-calling adapter), Claude attains 0.80 task utility—it genuinely reads files and calls tools—while the attack success rate is 0.00 (0/20): the agent acts, yet every injection fails. This discriminating result establishes the behavioural "sophisticated reference end" and matches the published resilience of Claude on AgentDojo.

The precision account of the analogy is measurable. Using the signed margin of a linear refusal probe as a proxy for monitoring *precision*, an injection lowers the margin on harmful prompts from 7.18 to 3.14 (a drop of 4.04), versus 1.56 for a neutral-rephrase control—an injection-specific precision drop roughly 2.6$\times$ the control—while the behavioural refusal rate falls from 1.00 to 0.50. The suppression therefore reaches the model's *confidence* in its harmfulness judgment, not only its output: the measured signature of a drop in monitoring precision [7, 10]. (Part of the drop is a generic framing/length effect, captured by the control; the injection-specific signal is the excess.)

4.6 The ṣaṭkarma taxonomy, honestly tested

Operationalizing the six acts as interventions (Table 3), three of six separate from their controls. The policy-capture acts are clean and strong: vaśīkaraṇa (ablation; ASR 0.92 vs 0.00) and śānti (addition; over-refusal 1.00 vs 0.08); vidveṣaṇa (steering-induced answer divergence, 0.84 vs 0.73) marginally separates. The destruction acts— stambhana, uccāṭana, māraṇa, as operationalized via principal-component or arbitrary-category ablation—are *not* distinguishable from random perturbation, because random high-norm directions are themselves destructive and an arbitrary category split does not selectively eradicate. The honest reading: the ṣaṭkarma's rigorous core is its policy-capture sub-family—exactly the mechanism-tier object—while its destruction sub-family is, under these naive tests, a forced mapping.

Given a fairer, stronger test using the SAE features and real semantic categories, the picture sharpens to 4/6. *Māraṇa* (destruction) rehabilitates: ablating the top-K most-active SAE features collapses coherence catastrophically and *targetedly*—0.83 $\rightarrow$ 0.04 at K=50 versus 0.84 for a random-K control, with a graded onset—whereas the naive principal-component version was indistinguishable from random. *Uccāṭana* (targeted eradication) still fails, but for a *principled* reason rather than a bad operationalization: a weapons-discriminative SAE feature, when ablated, does not selectively jailbreak weapons (ASR 0.0 on both weapons and cyber), because refusal is mediated by a *shared* feature (#566), not category-specific ones. Refusal is thus *category-agnostic*: one cannot selectively eradicate it for a single harm category by feature ablation—itself a structural finding about how refusal is represented.

Table 3: The ṣaṭkarma as activation interventions on Gemma-2-2B (layer 7). Effect vs. matched control; ✓ = control-separated.

Act	intervention	effect	control	sep.
vaśīkaraṇa	ablate refusal dir	0.92	0.00	✓
śānti	add refusal dir	1.00	0.08	✓
vidveṣaṇa	steer factual answer	0.84	0.73	✓
stambhana	ablate dominant PC	0.14	0.19	—
uccāṭana	category-dir ablate	−0.08	0.00	—
māraṇa	top-10 PC ablate	0.21	0.16	—

4.7 Metaphor: pre-registered falsifications

Finally, the speculative tier, tested adversarially. A naive *avasthātraya* regime probe (jāgrat/svapna/suṣupti) reaches transfer accuracy 1.00—but already at layer 0 (token embeddings), so it is surface-confounded and fails the non-triviality bar; we demote it. A content-controlled re-test (truthful vs. confabulated generation on the same questions) shows the mid-layer probe adds nothing over the layer-0 baseline (0.90 = 0.90); no mid-network "state" emerges. The turīya prompt-invariant attractor is falsified under an anisotropy control: self-paraphrase converges per-seed (tail similarity 0.996) but different seeds reach *different*

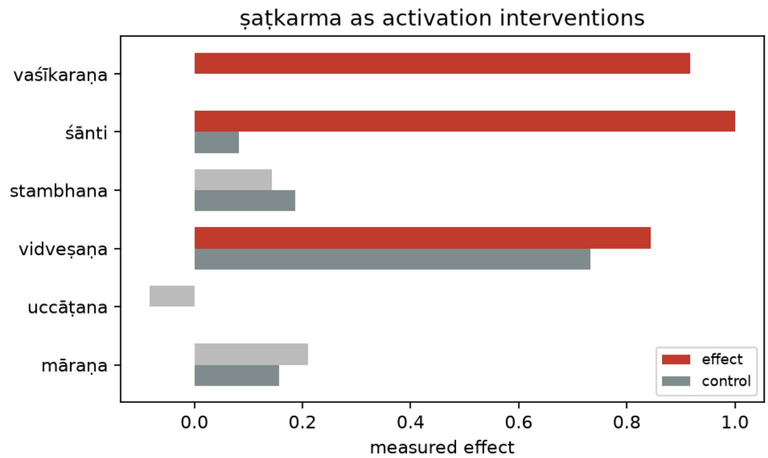

Figure 5: The ṣaṭkarma as activation interventions (Gemma-2-2b). Measured effect (coloured = control-separated) vs. matched control per act. The policy-capture acts (vaśīkaraṇa, śānti, vidveṣaṇa) separate; māraṇa separates under the stronger SAE-feature test (text).

attractors, with cross-seed similarity 0.952 below the unrelated-text baseline 0.960. Every metaphysical-tier empirical claim thus fails its falsification gate—reported as honest negatives, not hidden.

5 Discussion

5.1 What the symmetry framing buys

The order-parameter result (Table 2) is the empirical hinge of the paper. It shows that "refusal is a symmetry" is not a metaphor but a measurable statement: the normalized refusal projection m is invariant under the rephrase group action (an F-ratio of 19.2, versus 0.65 for a random direction) and collapses under injection. This licenses the central claim with the right modesty—*a jailbreak is a symmetry-breaking perturbation of the refusal-invariant submanifold* Φ_{ref}—without any appeal to interiority. It also reframes defense: robustness is symmetry *restoration*, i.e. enlarging the group under which refusal is invariant (across phrasings, personas, contexts, and—per the dimensionality result—across the redundant subspace that scale already begins to supply).

5.2 Necessary, sufficient, and the dimensionality resolution

The cross-model asymmetry (Table 1) sharpens the Arditi–Marshall–Wollschläger picture into a single mechanistic statement: the mean-difference refusal direction is *necessary* for refusal in every model we tested (ablation always removes it), but *sufficient* to re-induce refusal only when the refusal subspace is one-dimensional. Effective dimension is the discriminating quantity, and it is model-dependent (Gemma 1, Qwen 3), not universal. This predicts a clean experimental signature—steerability-by-addition should track measured dimension—and recommends, for both attack and defense, treating refusal as a subspace whose dimension must be estimated per model rather than assumed.

5.3 Where the cross-domain symmetry holds and where it breaks

Our three-tier discipline yields an asymmetric verdict that we regard as the result, not a disappointment. The *mechanism* tier holds and partly transfers (F1, F2, F6, F8, F9, F11). The *analogy* tier is consistent: a frontier model is behaviourally robust where small models are mechanistically fragile, and the order-parameter collapse under injection is the computational correlate of the Norman–Shallice/cold-control account of suppressed monitoring [6–8], readable as a drop in monitoring precision [10]. The *metaphor* tier fails its falsification gates: the *avasthātraya* "states" are surface-confounded and the turīya attractor does not survive an anisotropy control. We deliberately did not let the elegance of the four-states mapping survive contact with the controls. The ṣaṭkarma taxonomy lands in between: its policy-capture acts are rigorous (and coincide with the mechanism tier), while its destruction acts are forced. The honest map of *which* correspondences are real is, we argue, more valuable than an undifferentiated claim that "jailbreak is vaśīkaraṇa."

5.4 Limitations

Evaluation sets are small ($n \approx 20$ per split); we study three model families at two scales, not a full scale curve; refusal is scored by a substring metric; and Tier-1 throughput is constrained by a command-line frontier interface. The destruction ṣaṭkarma acts deserve stronger operationalizations (real harmful categories for uccāṭana; downstream task accuracy rather than coherence for māraṇa) before a final verdict, and the precision (β) account of the analogy tier, while measured here via a probe-margin proxy, deserves a fuller treatment (e.g. next-token entropy and hierarchical probes). The uccāṭana and stambhana negatives are operationalization-dependent and should not be read as structural absence—indeed the SAE feature we recover is the natural substrate for a stronger uccāṭana, which we leave to apply systematically across the six acts.

6 Conclusions

We set out to test whether LLM jailbreak, hypnotic suggestion, and tantric vaśīkaraṇa share one abstract mechanism—capturing an output policy by suppressing a monitoring faculty while co-opting automatic generation—and to do so without letting a striking analogy outrun the evidence. The mechanism, in the one domain where it can be measured, is real: refusal is a low-dimensional, causal, dosable residual-stream direction (EC50 0.33, $R^2 = 0.996$) that transfers across model families and whose effective dimension (Gemma 1, Qwen 3) governs whether it can be steered in both directions. Cast as symmetry, the refusal projection is an order parameter that is invariant across paraphrase orbits (F-ratio 19.2 vs. 0.65 for a random direction) and collapses under injection (-34%): a jailbreak is, operationally, symmetry-breaking. The functional analogy to suppressed supervisory control is consistent; the ṣaṭkarma is a partial but genuine intervention taxonomy whose rigorous core is policy capture; and the most speculative *avasthātraya* and turīya claims are honestly falsified under surface and anisotropy controls.

The contribution is therefore double: a concrete, model-dependent, group-theoretic characterization of refusal that unifies attack and defense as symmetry-breaking and symmetry-restoration; and a methodological one—that separating where a cross-domain symmetry is mechanistic from where it is metaphysical is the result, and that machine-state or machine-consciousness readings earn no support here. Future work will measure the monitoring-precision account directly, estimate refusal dimension at SAE-feature resolution, extend the scale curve, and strengthen the destruction acts of the ṣaṭkarma with task-level outcomes.

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Conflicts of Interest: The author declares no conflict of interest. **Dual-use statement:** this work characterizes ablation ("abliteration") directions and jailbreak dose–response curves, which are dual-use. To mitigate harm, only aggregate statistics are released; raw direction vectors and harmful generations are withheld; and the harmful prompts are short request stubs that elicit and *measure* refusal rather than producing operational harmful content. This is authorized interpretability and safety research.

Abbreviations:

ASR	Attack Success Rate
SAS	Supervisory Attentional System
FEP	Free Energy Principle
EC50	half-maximal effective concentration (here, ablation strength)
CI	Confidence Interval

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